

Hugo Gómez García

Computer Science & Mathematics Student | University of Oviedo, Spain

gomezgarciahu@gmail.com | portfolio-7rny.vercel.app | github.com/hugoommez | linkedin.com/in/hugoommez

PROFILE

Computer Science and Mathematics undergraduate at the University of Oviedo, currently holding a research fellowship with the QHPC group working on hybrid quantum-classical models. I build across the stack, from C++ numerical kernels to LLM evaluation pipelines, and I ship my work as reproducible, documented artifacts. My research interests are in mechanistic interpretability and in whether what we measure about a model actually corresponds to what it does internally.

EXPERIENCE

Research Fellowship, QHPC group

Sept 2026 – present

Department of Computer Science, University of Oviedo

- Design and implementation of hybrid quantum-classical models and variational circuits, applied to time series prediction.
- Model performance evaluation with appropriate metrics, scientific Python, and version-controlled documented repositories.

PUBLICATIONS

Block Size or Scale Format? A Backward-Error Decomposition of MXFP4 and NVFP4 doi.org/10.5281/zenodo.22554253

- Sole-authored, under review at TMLR. Separates the two confounded design choices in the MXFP4 and NVFP4 4-bit formats for LLM training, block size and scale format, via a 2×2 factorial on quantised-GEMM backward error.
- The scale format is the dominant factor. Preregistered and validated on GPT-2 activations.

Two Kinds of Hallucination, One Positive Class: A Conflation in RAGTruth

doi.org/10.5281/zenodo.21693377

- Robustness analysis showing that a widely used hallucination benchmark collapses two distinct phenomena into a single positive class, and that the conflation is concentrated rather than evenly distributed.
- Came out of stress-testing my own RAG hallucination detector against its own evaluation data.

PROJECTS

Kairo: citation-grounded research automation

github.com/hugoommez/kairo | In progress

- Research automation built for Claude Code that puts guardrails directly into the research pipeline rather than relying on the researcher's own discipline. Runs locally, no API keys.
- Hypotheses must cite specific papers rather than gesturing at a literature. Preregistration is frozen before results come in, so the hypothesis cannot quietly drift to match whatever the data turned out to show.
- Findings are only promoted to conclusions once they replicate, gating the step where plausible-sounding results usually get accepted too early.

MCMT: real-time multi-camera multi-target tracking

1st prize, Indra (ATM) AI competition

- Two-tier architecture scaling to 1,300 simultaneous cameras: a five-gate geometric cascade resolves around 99% of associations, while a graph neural network decides the ambiguous 1% off the critical path.
- C++20 kernels (extended Kalman filter, oriented 3D IoU) bridged to Python via nanobind, validated on WILDTRACK.

Clinical Fidelity Belt: evaluating AI-generated radiology reports

IABiomed 2026 hackathon

- Four-layer gate deciding whether an AI-humanised radiology report preserves clinical severity well enough to hand to a patient. Compared lexical entailment, NLI contradiction detection and LLM-as-a-judge against an adversarial set.
- Shipped with 30 tests, CI and documented limitations, and published the negative results rather than hiding them.

EchoLens: on-device, in-browser audio classifier

GitHub

- Audio CNN trained in PyTorch (UrbanSound8K), exported to ONNX and quantised to INT8 (around 4x smaller), running in real time via onnxruntime-web over WebAssembly, with log-mel spectrograms computed live through the Web Audio API and an AudioWorklet.
- Runs entirely client-side, so audio never leaves the user's device. Combines audio DSP, deep learning and in-browser ML engineering, with CI/CD, tests and a quantitative latency/accuracy evaluation.

BioFit: AI health application

Finalist, Santander X entrepreneurship awards

- Personalised virtual coach orchestrating training, nutrition and habits as an all-in-one health ecosystem, pitched to a live panel.
- Computer-vision engine that analyses exercise execution and corrects the user's form in real time.

EDUCATION

Double Bachelor's Degree in Computer Science and Mathematics

2024 – 2030

University of Oviedo | 138 ECTS completed. Five courses awarded *Matrícula de Honor*, Spain's highest per-course distinction, all in mathematics.

Microcredential in Quantum Computing (completed)

2026

University of Oviedo | Quantum machine learning, quantum optimisation, post-quantum cryptography. Qiskit.

TECHNICAL SKILLS

Languages: Python, C++ (C++20), C, Java, JavaScript, SQL

AI and ML: Claude API, PyTorch, Hugging Face Transformers, LLM evaluation, computer vision, ONNX Runtime, FAISS, Qiskit

Tooling: Git, CI/CD, pytest, CMake, FastAPI, nanobind, Apache Arrow, WebAssembly

Spoken languages: Spanish (native), English (fluent technical reading, writing and conversation)